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Manufacturing Processes

As a mechanical design engineer, I treat design for manufacturing as paramount: a design is only as good as the process that can produce it. I spent roughly 2.5 years as a manufacturing engineer at a premier plant for the world's leading metal-cutting tool company, and I machined parts and build my own prototypes in my former employer's shop. The hands-on habit started early — I grew up in the garage as the son of an engineer, and I owned and operated an LLC handyman business through high school and part of university. Whenever schedules and regulations allow, I personally build, test, and troubleshoot my prototypes end-to-end, including the labor, planning, and documentation (and, admittedly, the frequent troubleshooting).

The list below was compiled from memory and omits the most basic processes (hand drills, hammers, and the like). It covers processes I have personally performed, designed for, or witnessed in production.

Designed and physically used		Designed for and witnessed	Finishing, etc.
• Myriad controllers	• Manual mill	• DMLS, DLS, SLS, MJF/PJF	• Powder coating
• Any and every hand tool	• Manual lathe	• Casting (both die and investment)	• Painting
• Drilling	• Boring	• Injection molding	• Anodizing
• Reaming	• Wiring	• Sheet metal (bending, stamping, roll-forming)	• Thermoforming
• Angle grinder	• Deburring	• Thread forming	• Autoclaving
• Surface grinder	• Angle grinder	• Heat treatment	• Acetone bath
• Bench grinder	• Press brake	• Extrusions	• "Sand" blasting
• 3D printing (FDM, SLS, and SLA) [PLA, ABS, TPU, TPE, PETG, PACF, PC, Resin, Powder (PA12 and PA6)]		• Wire EDM	• Laser cutting
• Welding (mostly arc and stick)		• Piping	• Sanding
• CNC Machining (Programming and Running) ◦ Milling ◦ Turning ◦ Drilling ◦ Tapping ◦ Grinding		• Insulation	• Zinc Flake Coating
• CMM and other inspection		• Brazing	
• Hydraulic and arbor press			
• Most any form of cutting of most materials			